

RCS Solver Validation Report (Frequency Sweep)

Date: 2025-11-22

Project: RCS Cone Validation (Frequency Sweep)

Target: 1 m cone, 0.5 m base, 0.1 m tip radius

CPU Solver: MoM/PO - CPU Baseline

CPU Hardware: Xeon Gold (example)

GPU Solver: MoM/PO - GPU Accelerated

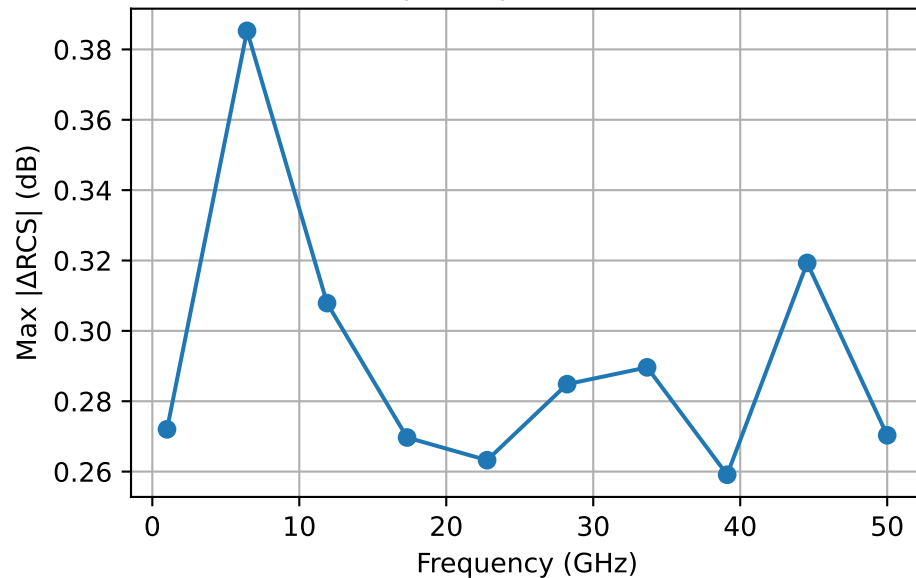
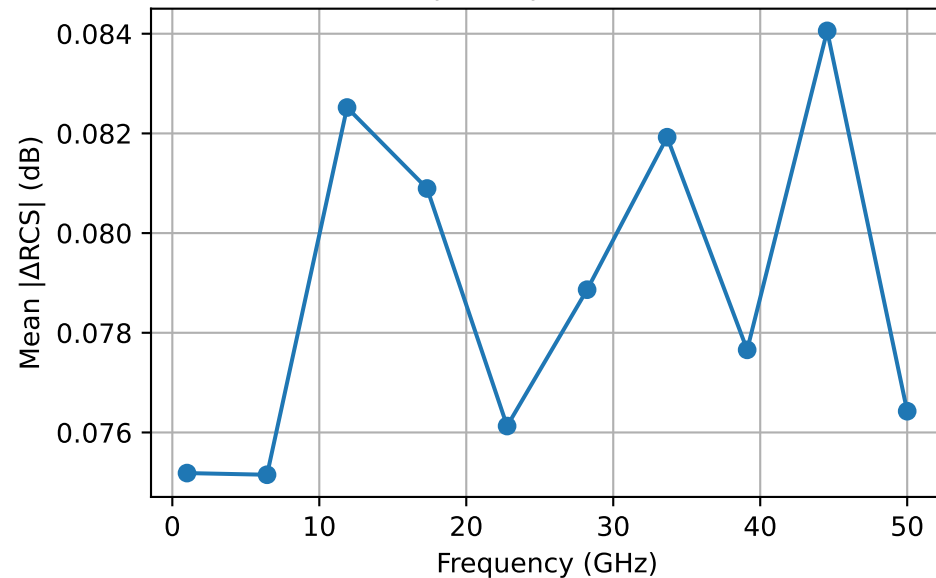
GPU Hardware: NVIDIA A100 (example)

Frequency range analyzed: 1.000 – 50.000 GHz

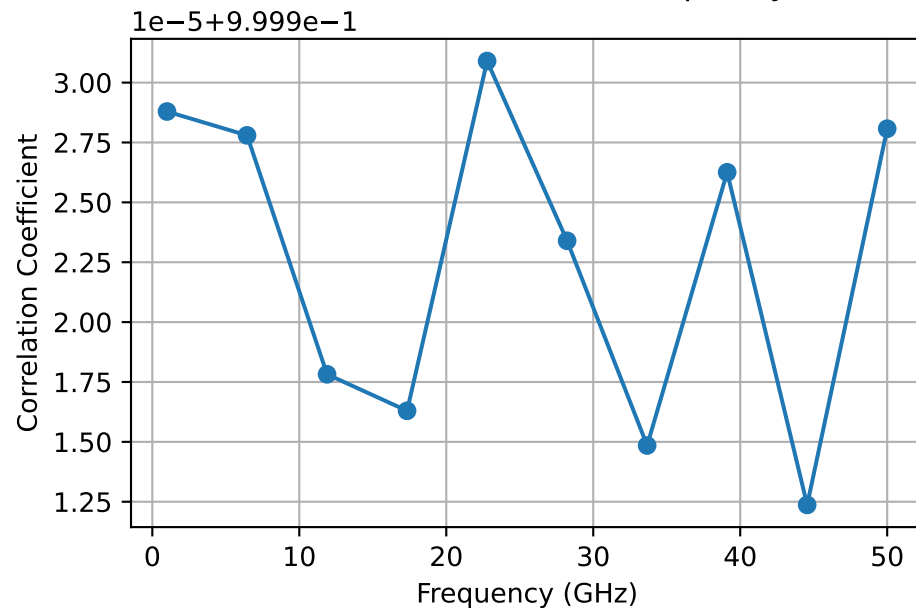
Polarization: HH

Global Summary of Key Metrics (across frequencies):

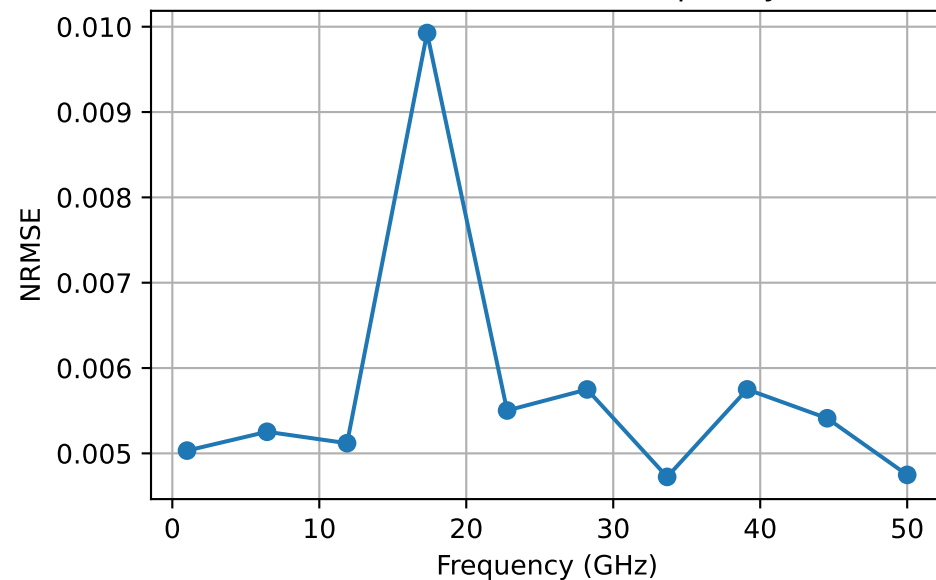
Max of Max $ \Delta\text{RCS} $	: 0.385 dB
Mean of Max $ \Delta\text{RCS} $	: 0.292 dB
Max of Mean $ \Delta\text{RCS} $	: 0.084 dB
Mean Pattern Corr.	: 0.99992
Min Pattern Corr.	: 0.99991

Max $|\Delta\text{RCS}|$ vs FrequencyMean $|\Delta\text{RCS}|$ vs Frequency

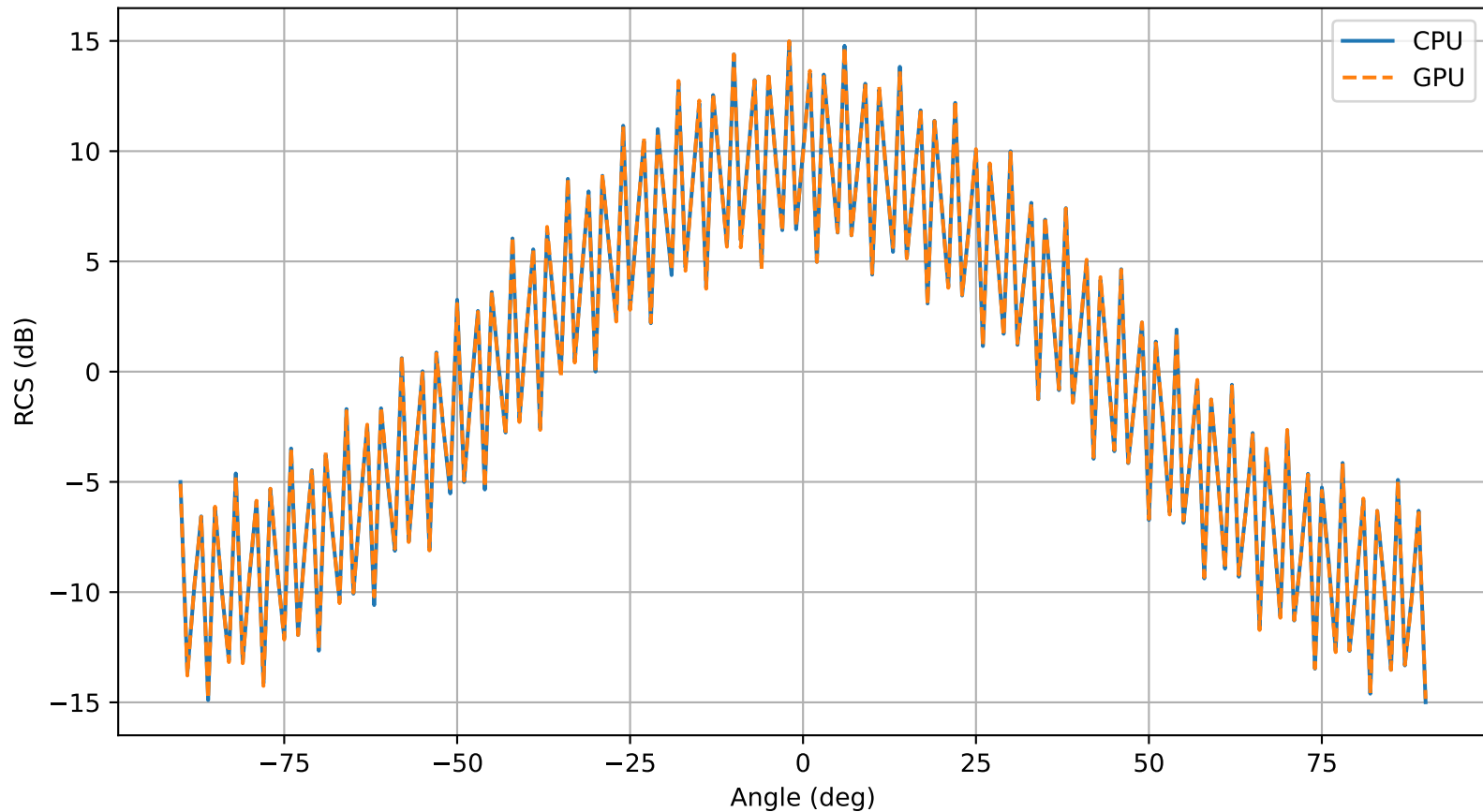
Pattern Correlation vs Frequency



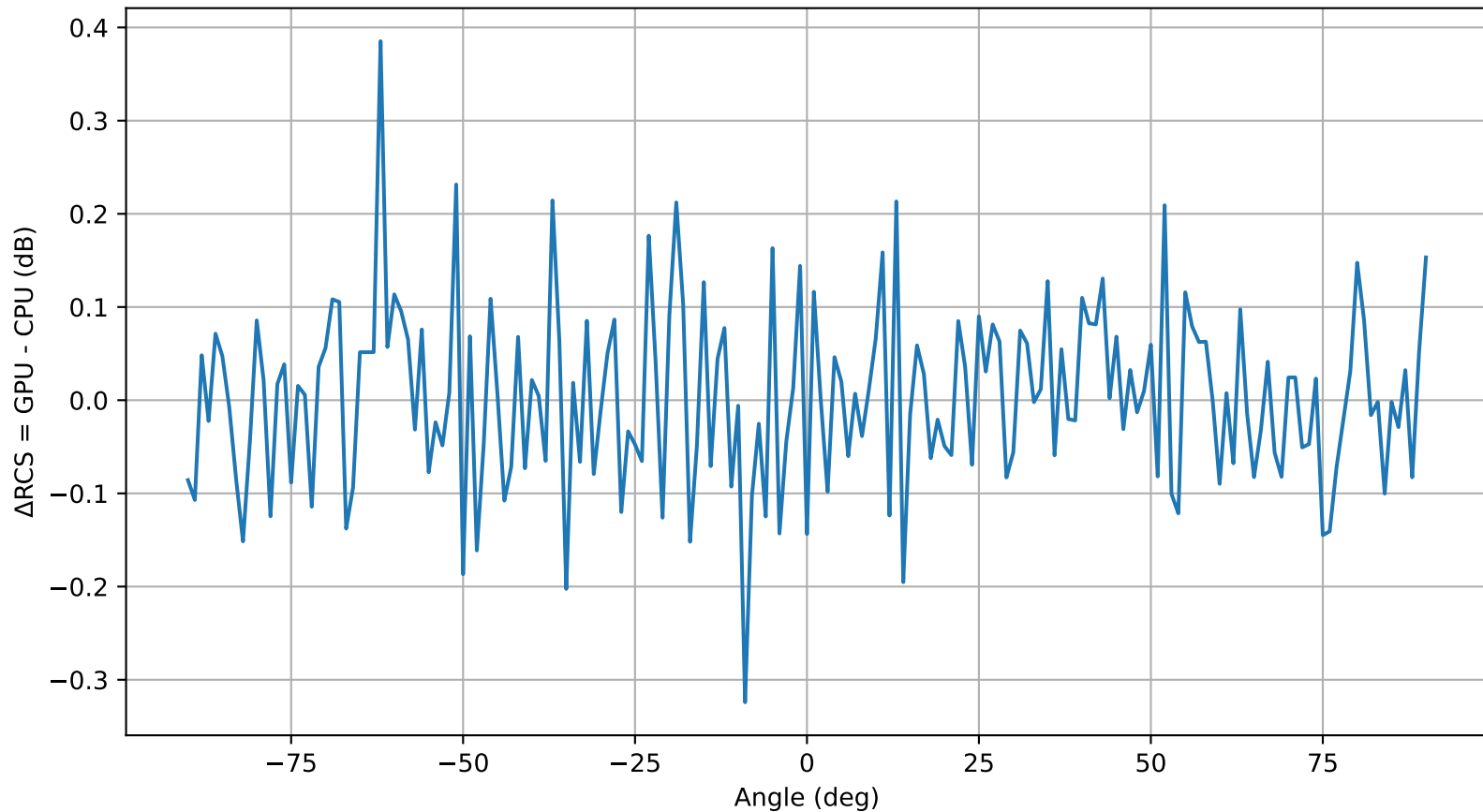
NRMSE (Linear) vs Frequency



CPU vs GPU RCS Comparison @ 6.444 GHz



Difference Plot (Δ RCS vs Angle) @ 6.444 GHz



Histogram of ΔRCS (GPU - CPU) @ 6.444 GHz

